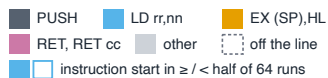
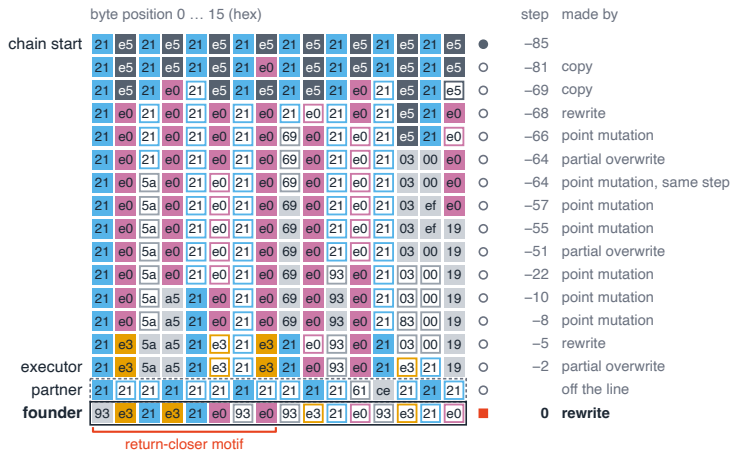


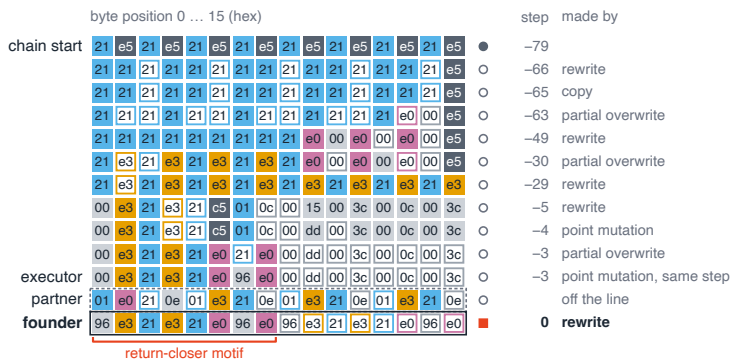
a

- open copier
- non-copier
- confined copier

World 31010: a program that copies neither its tape nor the code it runs writes the founder

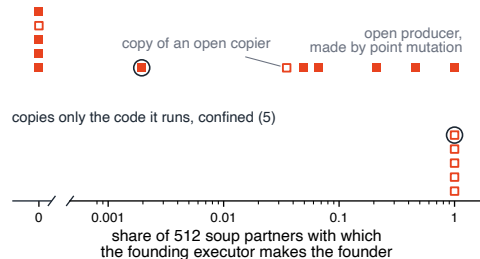
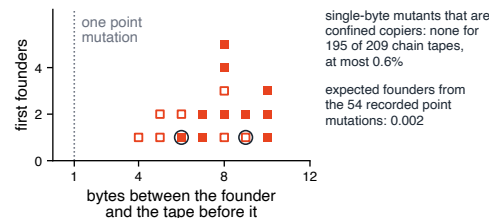
**b**

World 31019: a confined tape that copies only the code it runs arises by point mutation and writes the founder

**c**

- pointer entered the partner
- did not
- worlds in a, b

copies neither its tape nor the code it runs (12)

**d****e**